Chapter 6

SENSATION AND PERCEPTION

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Abstract: Sensory experiences and subjective perception serve as unique sources of tacit information that subconsciously influence people's cognition, emotions and behaviour. This chapter begins with an overview of key themes of sensation and perception in cognitive psychology, including tracing the history of perceptual processes, introducing different types of senses and perceptual processes and cognitivism-informed theories in this area. Following this, the discussion focuses on the application of sensation and perception within tourism field, discussing implications for research on sensescapes. This chapter concludes with an agenda for future research which delves into the role of multi-sensory and interaction of senses in touristscapes. **Keywords**: Sensation; perception; review; sensory experience; future research; embodiment

INTRODUCTION

The environment is multi-sensory in nature. Though visual sensations dominate, the accompanying sounds, smells, tastes and touch are all processed to create perceptions of the environment. The important role of the senses in shaping core human functioning is evident from ongoing discussions since the early days of philosophy (Aristotle, 2001; Plato, 2003). Recently, paradigm shifts in the scientific foundations of psychological research have prompted empirical investigations into the physiological underpinnings of

Cognitive Psychology and Tourism
Tourism Social Science Series, Volume 27, 59–72
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Published under exclusive licence by Emerald Publishing Limited
ISSN: 1571-5043/doi:10.1108/S1571-504320240000027006

human perception, highlighting the significance of the senses in shaping our understanding of the world (Krishna, 2012).

Sensation and perception are two separate processes that are very closely related. Sensation is input about the physical world obtained by the sensory receptors – it is biochemical and neurological in nature. Perception is the process by which the brain selects, organises and interprets these sensations (Krishna, 2012).

SENSATION AND PERCEPTION

The process by which the sensory organs – those involved with hearing and taste, for example – respond to external stimuli is called sensation. Sensation is deemed the foundation of how individuals interact with their surroundings, and thus a starting point for the process of deriving meaning from these interactions (Agapito, 2020). There are five types of sensation channels: vision, hearing, touch, olfaction and taste (Figure 6.1). Each sense organ requires a minimal amount of stimulation to detect a stimulus. This is called the absolute threshold.

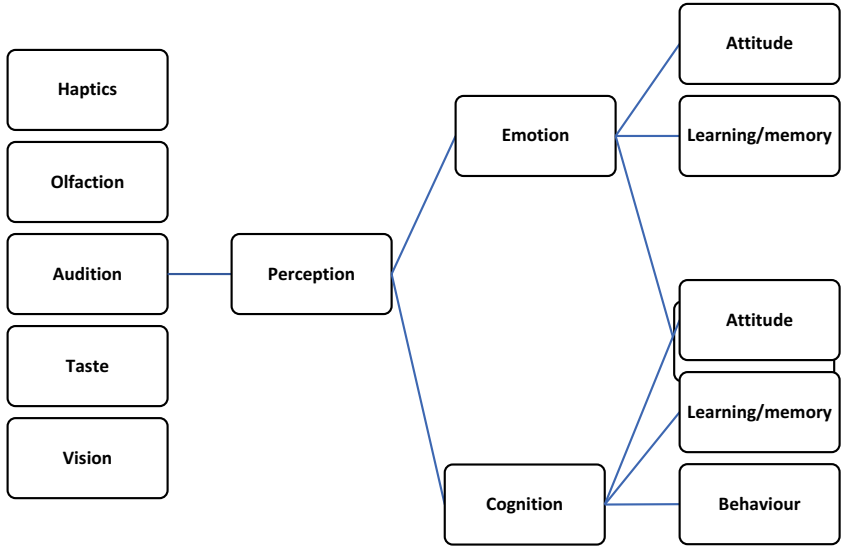


Figure 6.1. A Conceptual Framework of Sensory Processing.
Source: Adapted from Krishna (2012).

Sensations

Vision: Light enters the eye through the pupil, a tiny opening behind the cornea. The pupil regulates the amount of light entering the eye by contracting (getting smaller) in bright light and dilating (getting larger) in dimmer light. In the pupil, light passes through the lens, which focuses an image on a thin layer of cells in the back of the eye, called the retina. It is in the retina that light is transduced, or converted into electrical signals, by specialised cells called photoreceptors. Next, this electrical signal travels down the optic nerve.

After passing through the thalamus at the base of the brain, this signal makes its way to the primary visual cortex, where information about light orientation and movement begin to come together (Hubel & Wiesel, 1962). Information is then sent to a variety of different areas of the cortex for more complex processing. Some of these cortical regions are specialised for processing faces (fusiform face area) and body parts (extrastriate body area).

Hearing: People can get a large amount of information from the basic qualities of sound waves. The amplitude (or intensity) of a sound wave codes for the loudness of a stimulus; higher amplitude sound waves result in louder sounds. The pitch of a stimulus is coded in the frequency of a sound wave; higher frequency sounds are higher pitched.

Touch: Tactile stimuli – those that are associated with texture – are transduced by special receptors in the skin called mechanoreceptors. Just like photoreceptors in the eye and auditory hair cells in the ear, these allow for the conversion of sensations a form the brain can understand.

Olfaction (Smell): Both olfaction and gustation (taste) require the transduction of chemical stimuli into electrical signals. Unlike any of the other senses discussed so far, the receptors involved in the perception of both smell and taste bind directly with the stimuli they transduce. Odorants in the environment, very often mixtures of them, bind with olfactory receptors found in the olfactory epithelium. The binding of odorants to receptors is thought to be like how a lock and key operates, with different odorants binding to different specialised receptors based on their shape.

Taste: Taste works in a similar fashion to smell, only with receptors found in the taste buds of the tongue, called taste receptor cells. The binding of these chemicals with taste receptor cells results in the perception of the five basic tastes: sweet, sour, bitter, salty and umami (savory).

Some Senses Communicate Intrinsic Value. Sensations such as touch, smell and taste have an intrinsic valence, positive or negative, while others such as sight and hearing do not. The sensations that evolved later (for example, vision and hearing) do not contain intrinsic valence, although they are commonly labelled with valences generated from body states (Damasio &

Carvalho, 2013). Congruity theory highlights that individuals are more likely to develop positive attitudes towards products when the elements composing the consumption environment are perceived as consistent. Specifically, sensory studies show that when the arousal qualities of at least two sensory modalities are congruent, approach behaviours are improved (Agapito, 2020; Krishna, 2012).

Perception

The psychological process of making sense of the stimuli is called perception. It involves a series of steps involving what to perceive (what to pay attention to) and the interpretation of sensations. Both orientation (attention) to sensations and interpretation can be affected by goals. In addition, perceptions can also be affected by one's beliefs, values, prejudices, expectations and life experiences.

Amodal Information Processing. In the 1970s, the development of general models of information processing provided an integrative framework for the conceptualisation of human memory, judgement and decision-making (for authoritative contemporary treatments, see Lachman et al., 1979; Wyer, 1974). In a nutshell, the approach assumed that people acquire information through their senses and that this information is translated into a common code that is independent of the modality in which the information was initially acquired (Krishna & Schwarz, 2014). These amodal representations were assumed to be the building blocks of knowledge (semantic memory), on which the mind operates with general procedures that are also independent of the initial modality. Once stored, knowledge representations were assumed to be context independent, although context could influence how easily a stored representation could be accessed (Glenberg, 2010).

Multimodal perception: It is now known that experience is multimodel (vom Lehn, 2006). Real-world experiences typically involve multimodal sensations, wherein various senses combine to create a unified perceptual experience. Interestingly, one responds more strongly to multimodal stimuli compared to the sum of each single modality together, called the super additive effect of multisensory integration. The multimodal sensory stimuli are processed separately, and the results of those processes are qualitatively different from those of unimodal stimuli.

Top-down and Bottom-up Processing. While sensory receptors are constantly collecting information from the environment, it is ultimately how they interpret information that affects how one interacts with the world. Perception refers to the way sensory information is organised, interpreted and

consciously experienced. Perception involves both bottom-up and top-down processing. Stimuli have been defined as what is 'out there' in the environment. In other words, the external cues that stimulate the sensory receptors in turn are responsible for transmitting sensory information to the brain via electric signals. Top-down processing plays a role in perception via previous knowledge, memory and anticipation, which individuals bring to the situation (Goldstein, 2010). For example, current goals can direct attention to salient objects.

Mere Exposure Effect. When novel stimuli are repeated, people tend to rate the repeated stimuli as more likeable than nonrepeated stimuli (Glenberg, 2010). One well-supported explanation of the mere exposure effect is that repetition leads to more fluent processing, and that fluency leads to increased liking. In contrast, Lee (2001) provided an alternative discourse – uncertainty reduction explanation. Lee (2001) posits that individuals misattribute perceptual fluency to liking. The uncertainty reduction explanation posits that individuals prefer stimuli that are familiar and is consistent with findings that stimuli judged old were preferred to those judged new.

The Stroop Effect. The 'Stroop effect' is one of the best-known phenomena in cognitive science and indeed in psychology more broadly. It is also one of the longest standings, having been reported by Stroop (1935) in the published version of his dissertation. In its basic form, the task is to name the colour in which a word is printed, ignoring the word itself. When it is a colour word printed in a mismatched ink colour, this is very difficult to do and results in slow, error-prone responding (MacLeod, 1991).

Implicit Attitude and Implicit Association Test. Explicit attitudes are deliberate, intentional and available to conscious awareness. Implicit attitudes were defined by psychologists Greenwald and Banaji (1995, p. 8) as 'introspectively unidentified (or inaccurately identified) traces of past experience that mediate favorable or unfavorable feeling, thought, or action toward a social object [concept]'. Such attitudes can influence how humans see, think and react to the world, even without the individual having awareness of their existence, activity or influence (Nosek & Banaji, 2009).

Both the Stroop and implicit association tests are interference tasks. It was proposed by Greenwald et al. (1998) to measure individual differences in implicit social cognition. The test is a simultaneous classification task where participants have to classify pairs of stimuli into mutually exclusive categories. Participants' pre-existing associations between stimuli and categories can interfere with the classification task. As a result, responses are faster when the pre-existing associations match the pairing of categories on a

response key. The test scores reflect the interference effect from pre-existing associations on the classification task (Schimmack, 2021). As a result, the implicit association test and Stroop tests can indicate the existence of introspectively unidentified attitudes and hence aspects of schema.

Mind Body Connections

The human mind is dependent upon the body in its cognitive processing. Indeed, parts of the world different from those of the biological bodies can sometimes be processed as if they are part of them, as 'embodied' (Iriki et al., 1996). According to a more thorough formulation: given an external object X other than the biological body, 'X is embodied if some properties of X are processed in the same way as the properties of one's body' (De Vignemont, 2011, p. 82). Considering this definition, the 'embodiment' phenomenon would correspond to a specific modality of presentation of the properties of an object and would derive from the same way as the properties of the object are processed (Iani, 2021).

Embodiment discusses the profound connections between body and mind. Cultural geography and sensuous geography recognise the need to understand this mind and body interaction. From the perspective of cognitive science, it is known that bodily states may affect cognition, and that cognition may affect bodily states. Bodies provide sophisticated information-bearing channels, beyond vision or audition. A well-adapted mind uses any available, reliable signals. Similarly, mechanisms have evolved wherein mental states (such as fear) can affect physiological functions.

In embodiment cognition theory, sensory experiences and bodily feelings serve as unique sources of tacit information that subconsciously influence people's cognition, emotions and behaviour. This means sense-specific systems are involved in the cognitive processing which gives rise to 'embodied' mental representations. For instance, observing a given object (a bottle) will immediately activate the congruent motor information required to interact with it, such as a grasping movement and the appropriate hand shape (Goldinger et al., 2016).

The related term, 'situated cognition' indicates that cognitive processes change (either qualitatively or quantitatively) based on a person's goals and the immediate context (Goldinger et al., 2016). There are a number of approaches to embodiment and the theory has been used in a number of fields such as 'the feminist gendered body' (Alessandroni, 2017). In feminist literature, grounded cognition might mean that women think differently due to differences in male and female bodies. Embodiment has been related to aesthetic evaluation of stimuli such as dance (Kirsch et al., 2016; Papies et al., 2017).

Affordances: Affordance studies are often used in embodied theories. The concept of affordances originated in ecology, and can be seen as a way to combine the mind-body and environment (Buckley & Akhoundogli, 2019). The notion of affordance was first introduced by psychologist James Gibson (2015) to describe the potentialities for action afforded to individuals by the features of the environment. Affordances can be summarised as action possibilities offered from the environment to the animals that can seize these offering. A step affords stepability to an adult, not to a toddler (Pucillo & Cascini, 2014). This concept has been used to study service design, not from a psychological approach but more as a metaphor (Tomej & Xiang, 2020).

Grounded Cognition. The central theme of the grounded approach is that cognition emerges from interactions between classic cognitive processes: body and physical environment, and the social environment. Grounded cognition rejects traditional views that cognition is computation on amodal symbols in a modular system, independent of the brain's modal systems for perception, action and introspection. Instead, grounded cognition proposes that modal simulations, bodily states and situated action underlie cognition (Barsalou, 2008). Cognition is not a module in the brain that can be studied effectively in isolation but must be studied in the context of these other domains. Another central theme is that cognition supports situated action (Clark, 1998; Glenberg, 1997). Cognition is not an end but typically guides effective action in the world. Building on Barsalou's (2008) work, Krishna (2012, pp. 344–345) attempted to develop precise definitions of grounded cognition in terms of how it interacts with bodily states, situated actions and mental simulations, respectively.

Metaphors. For many consumer researchers, embodied cognition is probably synonymous with one particular stream of research, namely work that illuminates how sensory experience in one domain can influence cognition, emotion and behaviour in a different domain in ways that follow familiar metaphors (Krishna & Schwarz, 2014). Depending on theoretical perspective, metaphors are either seen as the unique driver of the observed effects (Lakoff & Johnson, 2008; Lakoff et al., 1999) or as expressions of the multimodal nature of mental representations (Anderson, 2010; Barsalou, 2008; Williams et al., 2009). The current interest in metaphors breaks with a long tradition in philosophy and linguistics that treated metaphors as rhetorical devices best reserved for poets (for a review, see Ortony, 1993). For example, the abstract concept of importance is closely associated with bodily sensations of weight (Schneider et al., 2011). In line with an embodied simulation account but contrary to a metaphor-enriched perspective, Schneider et al.'s (2011) study suggested that the connection between importance and weight goes beyond metaphorical mappings (see also Schema Chapter 12).

Presence. The construction of body awareness appears to be mediated by specific brain mechanisms involving a region of the cortex known as the temporoparietal junction. Damage to this brain region can generate distorted body awareness, such as feelings of a substantially elongated torso. Altered neural activity in this region through artificial stimulation can also produce an out-of-body experience, in which one feels like the body is in another location and one sees it and the world as if from the ceiling of the room. Remarkably, comparable brain mechanisms may also generate the normal awareness of the sense of self and the sensation of being inside a body. In the context of virtual reality, this sensation is known as presence (the compelling experience of actually being there). The normal localisation of the self may be equally artificial, in that it is not a given aspect of life but is constructed through a special brain mechanism.

Sensation and Perception in Tourism

Many of the tourism studies of senses accept that tourist experiences are intimately involved with their sensory perceptions of the destination. One experiences the surrounding landscape through the stimulation of senses (Rahman et al., 2017). Senses are deemed the foundation of how individuals interact with their surroundings, mediating the process of deriving meaning from these interactions (Agapito, 2020). A more complex idea of the tourism experience suggests that all five senses or a multisensory approach should be considered in, for example, gastronomic experiences and destination products (Pedro et al., 2020). All tourism destinations have unique and distinct sensory characteristics, which play an important role in creating memorable tourism experiences and tourist loyalty (Agapito et al., 2013; Pedro et al., 2020).

In the tourism context, Agapito et al. (2014) suggest that the sensory dimension of the tourist experience appears to be strongly linked to positive, unique and memorable experiences. They have therefore analysed the sensory dimension of the tourist experiences. In a rural context, they found that most references senses were to sight (26%), hearing (23%), taste (19%), smell (17%) and, finally, touch (15%). Each sense is, however, stimulated by different and specific environmental settings and provides different sensory information (Agapito et al., 2017). The entire sensory component (the multisensory stimulus) allows tourists to achieve an experience more truly meaningful and personalised (Kim & Fesenmaier, 2017a) and therefore more memorable. These personalised experiences suggest specific stimuli to provide memorable moments. Tourism experiences that include a multisensory offer may therefore contribute to the reinforcement of positive emotions, satisfaction and the creation of long-term memories and destination loyalty (Agapito et al., 2013; Diţoiu & Căruntu, 2014; Pawaskar & Goel, 2014). The

multisensory dimension of the tourist experiences has been termed a 'sense-scape'. Tourist experiences are not merely visual, but multisensory and tourism should involve various sensescapes, such as soundscapes, smell-scapes, tastescapes and the geography of touch in addition to landscapes (Liu et al., 2018).

Embodiment: In embodiment cognition theory, sensory experiences, and bodily feelings serve as unique sources of tacit information that subconsciously influence people's cognition, emotions and behaviour (Barsalou, 2008; Parrinello, 2012). One embodies other people's actions. When humans observe someone else executing an action, their brain covertly simulates the performance of the action being observed. In other words, the brain creates an internal representation of the observed motor programs that, in the absence of overt movements, is usually called 'action simulation'. Several studies have suggested that such a simulative mechanism is rapid and even obligatory (Iani, 2021). Embodiment may be seen as acting within the framework of mental simulations (Papies et al., 2017). These simulations draw upon multiple systems activated by a cue (Barsalou, 2020).

In the tourism field, Small et al. (2012) focuses on embodied experiences of disabled people. The paper concludes that the tourism industry and community must understand the multi-sensory nature of the tourist experience if quality accessible experiences are to be available for tourists with vision impairment. In addition, provision of multi-sensory experiences also enhances the experiences of sighted tourists. Sun and Lv (2021) discuss how a dark experience affects people. This is based on the idea that evaluative processing of general affective metaphors has been shown to bias perceptual judgements of stimuli. The mind-to-body pathway involves the dark experience exerting a measurable effect on participants' sensory expression in photographs and sketches. They then discover the body-to-mind pathway in which the expressions of darkness can be decoded by others. Xie and Sun's (2018) study expands the understanding of embodiment theory in dark tourism with implications for product design and marketing.

Embodied cognition effects on tourist behaviour (Kock & Ringberg, 2019). This notion is anchored in (bodily) feelings-as-information theory according to which people consult their affective states subconsciously to make judgements about objects (situations, people, the self or the environment) unrelated to the bodily senses (He et al., 2022). Traditional sensory marketing uses the smell of citrus fruit to signal freshness and cleanliness, whereas in embodied cognition, the sensory experience of the citrus smell is incidentally and metaphorically projected upon others as being morally purer. That is, incidental embodied experiences differ from tourism research which only examines the direct sensory dimensions of tourist experiences, such as sensory destination aspects. Kock and Ringberg (2019) provide numerous examples. Carrying weight makes hills seem steeper and distances

seem greater (Körner et al., 2015), highlighting the body's role in information processing.

Technological Embodiment. Although the integration between technology and a tourist's body is one of main features of technological embodiment (Tussyadiah et al., 2018), no previous studies have empirically tested this technological feature of virtual reality, nor analysed the mechanisms through which technological embodiment affects the customer experience. In the hotel industry, Flavián et al. (2021) study analyses the affective process by which technologies, with different levels of embodiment (virtual reality headsets, smartphones and desktop computers), can be effective in the hotel pre-experience stage. Tussyadiah et al.'s (2018) study provides a theoretical reflection on the phenomenon of embodiment relation in technological mediation and then assesses the embodiment of wearable augmented reality technology in a tourism attraction. The findings suggest that technological embodiment is a multidimensional construct consisting of ownership, location and agency. Flavián et al. (2019a) examined the effects of technological embodiment on customer engagement and behavioural intentions towards the destination through sensory simulation.

Landscapes: This dimension of sensescapes describes the visual channel and has attracted more scholarly attention than the other four human senses. Tourism studies have been systematically centred on the visual component of the tourist experience (Pan & Ryan, 2009), mainly focusing on the western-based sociological concept of tourist gaze developed by Urry (1990). The tourist gaze remains a key concept in tourism research, and its value choice is reflected in the destination choice and is reflected in the tourists' perceptions and behaviours, which form the destination's image (Samarathunga & Cheng, 2020). However, many studies using the gaze concept appear to assume that meaning lies within the object gazed upon, which is inconsistent with the cognitive approach adopted here (Zhang et al., 2022).

Smellscapes: An over-emphasis on the visual disregards that the unique character of a place can be imparted by its aromas (Dann & Jacobsen, 2003). Although traditional conceptions of the senses have often dismissed smell as vulgar, sub-human or lacking any aesthetic importance, some of the earliest anthropological studies recognised the importance of smell in understanding culture and social change (Śliwa & Riach, 2012). Olfactory sensations are extremely difficult to describe and categorise. Interviewees in Everett's food study particularly commented on the importance of smell in their appreciation of place, 'sometimes you can walk into a shop and walk past fruit, and you won't smell fruit, it's like, next! I think everything here is local produce, you can smell everything, you know' (2008, p. 344), thus supporting Dann and Jacobsen (2003) who insist that the senses must be considered in the

appreciation of the landscape, and that 'smellscapes' is a way of experiencing the city and countryside.

Soundscapes: Soundscapes are an important component of parks and outdoor recreation (He et al., 2018). The importance of sound to tourists lies in at least two aspects: one, sound can be the core element of tourist attractions (music) and it is an intangible resource that tourists consume when travelling; and, two, sound helps tourists to establish the sense of a place and enhances their perception and understanding of a natural destination. The unique soundscape of destination is not only a key part of its attraction but also an element that can enrich the tourism experience (Kankhuni & Ngwira, 2022). Jensen (2016) noted that sounds in tourism should be revalued and researched by utilising the localised, sensuous and embodied act of listening.

Tastescapes: Much research on smell and memory has been done within the basic sciences, psychology and recently in marketing (Krishna, 2012). Tastescapes have been associated with the field of food tourism (Everett, 2019), but little attention paid to this issue from cognitive psychology. Everett (2019) deploys the term tastescape as an entry point to delineate the evolution of food tourism research. Inspired by the idea of a therapeutic landscape, Su and Zhang (2022) propose the notion of tastescapes to highlight that well-being is a multisensory condition, and tea tasting can be conducive to well-being and health.

Touchscapes: Personality analogies often dwell on haptic characteristics (Krishna, 2012). In studies of altruistic behaviour, Peck and Wiggins (2006) gave people brochures for an arboretum that either had a 'touch' element associated with it (a tree bark or feather in the front page of the brochure) or did not. They found that people's willingness to donate to the arboretum increased with a touch element in the brochure for people who had a high autotelic need for touch. People who have a high autotelic need for touch engage in touch because it is fun, interesting or enjoyable, an experience that is more hedonic than instrumental. There was no difference in donation willingness with and without the touch element for those who had a low autotelic need for touch.

Perception of Time. The sense of time has no direct sensory system that specifically supports one's appreciation of the duration of events and activities (Pearce, 2020). Nevertheless, it is widely recognised that people do have skills in assessing how much time has passed, even if one does not fully understand how the relevant information is processed. In a tourism setting, how tourists perceive, use and respond to time in their holiday experiences provide opportunities for recasting the design of many experiences (Pearce, 2020). Li et al.'s (2022) recent study postulates time perception is an inevitable factor influencing tourist behavioural decisions during the course of

travel; however, their study is based on social psychology. Instead, drawing on work in cognitive psychology and neuroscience, Pearce (2020) considers ideas about the perception of time and temporal features of tourists' social episodes and articulates six principles for furthering the consideration of time in the better design of contemporary tourism experiences. These are specifying duration, managing, and filling downtime, preparing for emergencies, tailoring tourist sequences, encouraging engagement and implicating time in memorability.

Sensory Marketing and Design

Krishna defines 'sensory marketing' as 'marketing that engages the consumers' senses and affects their perception, judgment, and behavior' (2012, p. 332). From a practical standpoint, sensory marketing may trigger subconscious cues which tourist perceptions of abstract notions of the product (its sophistication or quality) are sensitive to. Despite the widespread popularity of explicit marketing appeals to consumers, implicit subconscious cues have untapped potential in influencing consumers attitude and behaviour. These sensory triggers may result in a consumer's recognition of (desirable) brand attributes, different from those provided in advertising. An understanding of sensation and perception as it applies to consumer behaviour may help to understand these sensory triggers (Krishna, 2012; Krishna & Schwarz, 2014).

Digital sensory marketing has emerged as information and communication technologies have evolved (Petit et al., 2019). Integrating new technologies into multisensory online experience, other interfaces involving sensory triggers (including touch screens, together with a range of virtual, and augmented solutions) are increasingly salient and available to people when interacting online (Petit et al., 2019). With the advancement of the field of human–computer interaction, online environments are likely to engage more of the senses and be more connected with offline environments in the near future. This expansion will likely coincide with an increasing engagement with the consumer's more emotional senses, namely touch/haptics, and possibly even olfaction (Agapito et al., 2013). Forward-thinking marketers and researchers are thus expected to appropriate the emerging techniques to deliver richer online experiences for tomorrow's consumers (Petit et al., 2019).

Sensory Design. The importance of addressing multisensory aspects in the design of tourism experiences and places has become apparent in a recent strand of research (Agapito, 2020). Recent comprehensive action-oriented frameworks in the book of *Design Science in Tourism* (Fesenmaier & Xiang, 2017) involve a sustainable approach to tourism experiences and place design, embraces the senses as one of its foundational aspects. They conclude experience as a response to the environment is generally considered a

systematic process where sensation is the first order of experience. How people 'interpret' the stimuli and 'make' meaning from them and generally is framed within the mechanisms of perception. Thus, it is argued, travelers' perceived senses are not direct records of the world around them. Rather, they are 'constructed' internally along with constraints imposed by the construction of the nervous system and its functional abilities (Fesenmaier & Xiang, 2017).

Affordances in tourism: There is limited research discussing the application of affordances within the area of tourism design (Tomej & Xiang, 2020). Recently, Fesenmaier and Xiang (2017) listed affordance as one of the six components of the 'experience production system'. However, it appears that the concept has not yet received a careful, systematic examination in terms of how it is connected to various aspects of design in tourism. Tomej and Xiang (2020) present a case study wherein they used the concept of affordance to analyse the service experiences in an organised holiday for a group of tourists with and without visual impairment.

CONCLUSION

Literature reviews are available on specific senses, on using senses for marketing purposes, or on topics such as wine (Agapito, 2020; Brochado et al., 2021). This chapter provides a systematic overview of research incorporating the sensory component of tourism from a cognitive psychology perspective. Agapito's (2020) review finds that regardless of the variety of research methods used and multidisciplinary approaches in the research addressing the sensory component of the tourism experience, it can be observed that the articles in the dataset are predominantly developed from the perspective of tourists. Moreover, in general, only one stage of the tourism experience (pre-visit, onsite or post-visit) is addressed. A few exceptions (Agapito et al., 2017; Ballantyne et al., 2011) perform follow-up research. Notwithstanding, it is acknowledged that future research seeking to inform more effective tourism experience design around the senses should analyse how sensory-informed experiences evolve across different stages of the tourism experience and involve different stakeholders.

As in other chapters, the study of the senses has been undertaken from a variety of disciplinary perspectives, the gaze concept from sociology. These studies appear to assume that meaning lies within the object gazed upon, which is inconsistent with the cognitive approach adopted here. Despite this, there may be some benefit to reexamine the concept of the gaze from a cognitive psychology perspective. This may help, for example, to understand the consequences of 'gazing' such as an inability to engage with the viewed surroundings. Viewed from a cognitive perspective, 'gazing' involves

difficulty in interpreting unfamiliar stimuli due to a lack of cues and past experience. This would appear to be a useful focus of further enquiry.

By addressing contemporary technological developments, neurophysiological instruments, multisensory components can be cannibalised by technology to enhance tourism experiences in an inclusive manner (Flavián et al., 2019b). However, the application in tourism remains limited. There is a surge of interest among a body of researchers who focus on the effectiveness of virtual reality on advancing understanding of sensory components. Alongside the advancement of immersive technologies, an analysis of the research methods employed in the articles indicates that sensory data collection has primarily been conducted through questionnaires and interviews. These methods could be complemented by using physiological measurement devices and mobile devices to better understand how senses interact with each other, and how different profiles of individuals react to different stimuli in diverse tourism contexts (Kim & Fesenmaier, 2017b). Technology can also aid the design of experimental research and longitudinal studies to examine the evolution of sensory-informed experiences across different stages of the tourism experience.

While a lot of work which predominantly focuses on single channel of five senses of humans has been done on sensory marketing in the last two decades, there is still a great need for additional research on the interaction of multi-senses: sensory congruence, sensory conflict and synaesthesia. In a retail setting, the congruence between the arousing qualities of scent and music improves approach behaviours (Mattila & Wirtz, 2001). Krishna et al. (2010) focus on the effect of the congruence between semantic associations of smell and touch. In contrast, many other studies have also examined vision–touch discrepancy and found strong or complete dominance of vision over touch. Little tourism work has examined sensory conflict even though many sensory conflicts exist within the consumption domain, which is an area for future research (Krishna, 2012).

Finally, synaesthesia, a new theoretical framework for understanding intersensory development, has been used to describe that sensory stimuli induce extra cross-modal or intramodal percepts (Spector & Maurer, 2009). Recent findings from cognitive psychology, functional brain imaging and electrophysiology have shed considerable light on the nature of synaesthesia and its neurocognitive underpinnings, which warrants scholarly attention from tourism (Grossenbacher & Lovelace, 2001).